

Assignment-2 Report

1. Number of nodes- 180
2. Number of edges - 424
3. Average node degree – 4.71
4. Average local clustering coefficient- 0.433
5. PPI enrichment p-value- $< 1.0e-16$

Methodological Justification for Layout Selection

The global protein-protein interaction (PPI) network was imported into Cytoscape and optimized using the **Prefuse Force-Directed Layout algorithm**. This physics-based simulation models proteins as physical objects carrying an electrostatic charge that repels each other, while treating interaction edges as biological springs that pull connected nodes closer together.

This approach is highly appropriate for therapeutic target prioritization because it naturally pulls highly connected "master regulator" hubs into prominent central focal points. Concurrently, it automatically groups tightly interacting proteins into distinct spatial clusters, visually accentuating the core functional disease modules while minimizing overlapping edge lines.

Prioritized Molecular Target Panel

Network topology was calculated using Cytoscape's native *NetworkAnalyzer* tool under an undirected network model [1, 2]. To identify the core driving engine of cervical carcinoma, nodes were ranked by their Degree Centrality (total direct functional connections). The top 5 highest-degree molecular hubs are documented in the table below:

Gene Symbol	Degree Centrality	Biological Function & Role in Cervical Cancer
CDK1	31	Cyclin Dependent Kinase 1: The ultimate driver of the cell cycle; forces

		cancer cells to divide uncontrollably.
CDC20	26	Cell Division Cycle 20: Activates the anaphase-promoting complex, causing rapid chromosomal separation during mitosis
TOP2A	26	DNA Topoisomerase II Alpha: Controls DNA structure during replication; highly active in rapidly dividing tumors.
CCNB1	25	Cyclin B1: Binds to CDK1 to trigger mitosis entry; a major biomarker for poor clinical survival.
RRM2	24	Ribonucleotide Reductase Regulatory Subunit M2: Supplies the raw nucleotides needed for fast tumor DNA synthesis.

Module Rank	MCODE Score	Nodes	Edges	Seed Node
Cluster-1	18.667	19	168	ZWINT
Cluster-2	5.667	7	17	CDC7
Cluster-3	5.143	8	18	DSG3

Module	Gene	Dominant GO Biological Process
Module-1	CDK1, CDC20, CCNB1, TOP2A	Mitotic Nuclear Division, Chromosome Segregation, Spindle Assembly
Module-2	MCM2, MCM5, MCM6, GMNN	DNA Replication Initiation, Pre-Replicative Complex Assembly

Report Question

1.Does the observed interaction density suggest random co-occurrence or coordinated biological organization? Support your answer using STRING statistics.

Answer-

The observed interaction density provides definitive statistical evidence of **coordinated biological organization** rather than random co-occurrence.

Justification:

- **PPI Enrichment p-value** : This is the most critical metric. A p-value this tiny means the probability of these connections happening by pure random chance is essentially zero. Our 180 proteins are mathematically proven to be functionally linked together.
- **High Connection Density**: Our network contains **180 nodes** and **424 edges**. This high number of edges indicates that our proteins are heavily interacting with one another to execute cellular tasks, which would not happen in a scrambled or random set of genes.
- **Average Node Degree (4.71)**: On average, every single protein in our network connects directly to nearly 5 other proteins. This shows a highly cohesive communication web rather than isolated, independent molecules.
- **Average Local Clustering Coefficient (0.433)**: This score proves that Our network is highly modular. Instead of being spread out thinly, the nodes are tightly packed into distinct, collaborative functional groups or "neighborhoods" that work together to drive the disease.

2. If a protein is significantly differentially expressed but absent from every detected network module, should it still be considered a therapeutic target? Justify using principles of network biology

Ans-

Yes, it should still be considered a therapeutic target, but its role and priority will be different from the core module targets.

Justification:

- **Functional Autonomy:** Network modules represent tightly packed, dense protein factories (like a cell division machine). However, some vital proteins work alone or act as simple "on/off" switches rather than part of a large complex.
- **Transient Interactions:** A protein might be highly active but only touches other proteins for a brief second (transient signaling). These brief touches do not show up as permanent, dense clusters in a static network map, but the protein remains a critical driver of the disease.
- **Input/Output Bridges:** These proteins often sit on the outer paths of the network. They act as the initial sensors that pick up a signal and pass it into the main modules. Knocking them out can cut off the signal before it ever reaches the rest of the network.

3. Assume the highest-ranked consensus protein is experimentally inhibited. Explain three different mechanisms by which the disease network could compensate for the loss of that protein.

Ans-

If we experimentally shut down our top consensus target (**CDK1**), the disease network can fight back and survive using these three network biology mechanisms:

- **Mechanism 1: Alternative Routing (Path Redundancy)**
Networks have backup pathways. If CDK1 is blocked, the cell can reroute its signals through parallel kinase paths (like CDK2 or CDK4). The signal bypasses the blocked CDK1 roadblock to keep the disease growing.
- **Mechanism 2: Network Robustness via Hub Clustering**
Our results showed a massive 10-protein cluster shared by MCC, Degree, and MNC (like CCNB1, TOP2A, and MELK). Because these other hubs are so highly connected to each other, they can share the network workload and keep the overall module active even when CDK1 is missing.

- **Mechanism 3: Feedback Loops and Upregulation**

The network can sense a drop in downstream signaling. It will trigger negative feedback loops that force the cell to create even more upstream signaling molecules (like cyclins). This flood of signals can overactivate the remaining network nodes to bypass the drug block entirely.

4. The pharmaceutical company can only validate two proteins experimentally. Describe a rational framework for selecting two candidates from the consensus target panel. Your answer must integrate:

☐ **Network topology** ☐ **Functional enrichment** ☐ **Biological plausibility** ☐ **Druggability**

Ans-

Since we only have one single master consensus protein (**CDK1**), a rational framework must look at your next top-tier network tier—the **10-protein network hub cluster**—to pick up the second ideal candidate.

- **1. Network Topology:**

We select **CDK1** as candidate #1 because it scored perfectly across all five topological filters. For candidate #2, we pick a protein from the core 10-hub cluster (like **TOP2A** or **CCNB1**) that scored highest in the direct neighborhood metrics (MCC, Degree, and MNC).

- **2. Functional Enrichment:**

We check functional enrichment results (like Gene Ontology or KEGG paths). We Pick two proteins that control different, essential biological weapons of the disease. For example, we pair a cell cycle engine (CDK1) with a DNA replication/repair protein (TOP2A). This hits the disease from two different angles.

- **3. Biological Plausibility:**

We look at existing medical literature to ensure both proteins are highly active in the specific disease tissue we are studying. Their known biological jobs must match perfectly with the aggressive traits of the disease.

- **4. Druggability:**

We check public databases (like Drug Bank or ChEMBL) to see if small-molecule inhibitors already exist for these targets. Choosing proteins with known chemical binding pockets makes moving them into actual lab testing much cheaper, faster, and more likely to succeed.

